

Announcements:

(from HW3 on)

- All HW pushed back 1 week

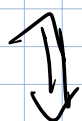
- We may or may not have HW6, TBD.

Multiple Linear Regression Model

There are some variations on how to specify the MLR model (e.g. index order, # of predictors as p or $p-1$).

In order to all be on the same page, let's all use:

$$y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_{p-1} X_{(p-1)i} + \varepsilon_i \quad i=1, \dots, n$$



$$Y = X\beta + \varepsilon$$

where

this class' notation

$$X = \begin{bmatrix} 1 & X_{11} & X_{21} & \dots & X_{(p-1)1} \\ 1 & X_{12} & X_{22} & & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{1n} & X_{2n} & \dots & X_{(p-1)n} \end{bmatrix}$$

$\times$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{pmatrix}$$

LS Est was:

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} Q(\beta) = \underset{\beta}{\operatorname{argmin}} (y - X\beta)^T (y - X\beta)$$

$$= (X^T X)^{-1} X^T y \quad \text{assuming } X \text{ has full rank.}$$

Fitted Values:

$$\hat{y} = X \hat{\beta} = X \underbrace{(X^T X)^{-1} X^T}_H y$$

$$= H y$$

$n \times n$

Claim: H is a projection matrix

① symmetric & ② idempotent

① Is $H = H^T$?

yes!
Pf:

$$\begin{aligned} H^T &= (X(X^T X)^{-1} X^T)^T \\ &= (X^T)^T ((X^T X)^{-1})^T X^T \\ &= X (X^T X)^{-1} X^T \\ &= X (X^T X)^{-1} X^T = H. \end{aligned}$$

$$(ABC)^T = C^T B^T A^T$$

$$AA^{-1} = I$$

$$(AA^{-1})^T = I^T = I$$

$$\Rightarrow (A^{-1})^T A^T = I$$

$$(A^{-1})^T \cancel{(A^T)} \cancel{(A^T)^{-1}} = (A^T)^{-1}$$

② idempotent

Claim:

$$H^2 = H$$

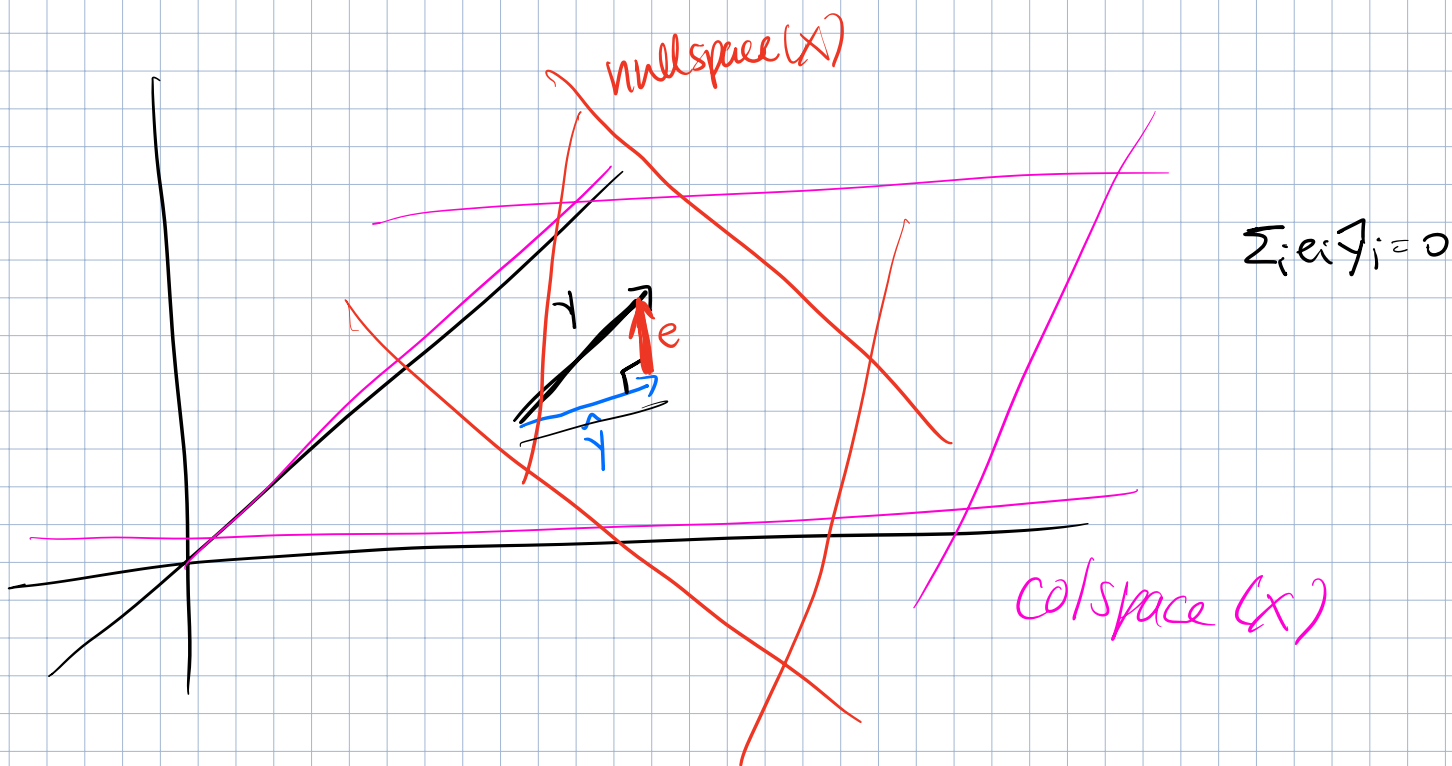
Pf:

$$H^2 = H \cdot H = \overbrace{(X(X^T X)^{-1} X^T)}^H \overbrace{(X(X^T X)^{-1} X^T)}^H$$

$$= X(X^T X)^{-1} \underline{\underline{X^T X}} (X^T X)^{-1} X^T$$

$$= X(X^T X)^{-1} I \cdot X^T$$

$$= \underline{\underline{X(X^T X)^{-1} X^T}} = H.$$



Residuals:

$$e_i = y_i - \hat{y}_i$$

$$e = y - \hat{y} = y - Hy = \underline{(I - H)y}$$

Is the act of multiplying by $(I - H)$ also a projection?

Maybe, let's try to prove

$I - H$ is ① sym.
 ② idempotent.

Pf:

$$① (I - H)^T = (I^T - H^T) = I - H. \quad \checkmark$$

$$\begin{aligned} ② (I - H)^2 &= \overbrace{(I - H)(I - H)} \\ &= (I - IH - HI + H^2) \\ &= I - H - \cancel{H} + \cancel{H} = I - H \end{aligned}$$

How to estimate σ^2 ?

$$\boxed{\hat{\sigma}^2 = \frac{SSE}{n - p}}$$

Why?

$$\begin{aligned}
 SSE &= \sum_i e_i^2 = e^T e \\
 &= (I-H)Y^T (I-H)Y \\
 &= Y^T (I-H)^T (I-H) Y \\
 &= Y^T (I-H) Y
 \end{aligned}$$

MLR:

$$Y \sim N(\underline{X}\underline{\beta}, \sigma^2 I)$$

$$\Rightarrow e = (I-H)Y \sim N(\underline{0}, \sigma^2(I-H))$$

$$\begin{aligned}
 E((I-H)Y) &= (I-H)E(Y) = (I-H)\widehat{X}\beta \\
 &= (IX - HX)\beta \\
 &= (X - X(X^T X)^{-1} X^T X)\beta \\
 &= (X - X)\beta = \underline{0}
 \end{aligned}$$

$$\text{Var}(e) = \text{Var}((I-H)Y) = (I-H)\text{Var}(Y)(I-H)^T$$

$$= (I-H) \sigma^2 I (I-H)$$

$$= \sigma^2 (I-H)$$

$$\text{Var}(AY) = A \text{Var}(Y) A^T$$

$$\rightarrow e = (I - H)Y \sim N(\underline{0}, \underline{\sigma^2(I - H)})$$

$$\rightarrow \frac{(I - H)Y}{\sigma} \sim N(0, (I - H))$$

$$\frac{SSE}{\sigma^2} = \frac{e^T e}{\sigma^2} = \frac{((I - H)Y)^T (I - H)Y}{\sigma^2} = \left\| \frac{(I - H)Y}{\sigma} \right\|^2$$

[What is $\|a\| := \sqrt{\underline{a_1^2} + \underline{a_2^2} + \dots + \underline{a_n^2}}$ ← Norm/length of a]

$$(I - H)^{-1/2} \frac{(I - H)Y}{\sigma} \sim N(0, \cancel{(I - H)^{1/2}} (I - H) \cancel{(I - H)^{1/2}})$$

$$\parallel N(0, I)$$

$$\frac{(I - H)^{1/2} Y}{\sigma}$$

(property of quad. forms of normal rv)

$$\left\| \frac{(I - H)^{1/2} Y}{\sigma} \right\|^2 \sim \underline{\underline{\chi^2_{df}}} \quad \text{where } \underline{\underline{df = rank(I - H)}}$$

$$\underline{\underline{\frac{Y^T (I - H) Y}{\sigma^2}}}$$

What is the rank of $I-H$?

for idempotent matrices, the rank = the trace.

$$\begin{aligned}\text{rank}(I-H) &= \text{tr}(I-H) = \text{tr}(I) - \text{tr}(H) \\ &= n - \text{tr}(X(X^T X)^{-1} X^T)\end{aligned}$$

$$= n - \text{tr}(\underbrace{(X^T X)^{-1}}_{p \times p} \underbrace{X^T X}_{p \times n \times n \times p})$$

$$= n - \text{tr}(I_p)$$

$$= n - p.$$

Cyclic

$$\begin{aligned}\text{tr}(ABC) &= \text{tr}(BCA) \\ &= \text{tr}(CAB) \\ &= \text{tr}(ABC)\end{aligned}$$

$$\frac{SSE}{\sigma^2} = \left\| \frac{eTe}{\sigma} \right\|^2 = \left\| \frac{(I-H)y}{\sigma} \right\|^2 \sim \chi^2_{n-p}$$

$$E\left(\frac{SSE}{\sigma^2}\right) = n-p$$

$$\frac{1}{\sigma^2} E(SSE) = n-p$$

$$\frac{1}{n-p} E(SSE) = \sigma^2 \Rightarrow E\left(\underbrace{\frac{SSE}{n-p}}_{\hat{\sigma}^2}\right) = \sigma^2$$

$\hat{\sigma}^2 = \frac{eTe}{n-p}$ is unbiased for σ^2 .

Distributions of $\hat{\beta}$ & $\hat{y}$:

$$\hat{\beta} = \underbrace{(X^T X)^{-1}}_A X^T y$$

Recall

$$y \sim N(\mu, \Sigma)$$

$$Ay \sim N(A\mu, \underline{A\Sigma A^T})$$

So under the MLR model:

$$y \sim N(X\beta, \sigma^2 I)$$

$$\hat{\beta} = (X^T X)^{-1} X^T y \sim N(\cancel{(X^T X)^{-1} X^T} X\beta, (X^T X)^{-1} X^T (\sigma^2 I) (X^T X)^{-1})$$

$$= N(\beta, (X^T X)^{-1} X^T \sigma^2 I (X) (X^T X)^{-1})$$

$$= N(\beta, \sigma^2 (X^T X)^{-1} \cancel{X^T X} (X^T X)^{-1})$$

$$= N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\therefore E(\hat{\beta}) = \beta$$

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Dist of $\hat{y}$?

$$y \sim N(X\beta, \sigma^2 I)$$

$$\hat{y} = Hy \sim N(HX\beta, (H)(\sigma^2 I)(H^T))$$

$$N(X\beta, \sigma^2 H)$$

$$HX\beta = (X(X^T X)^T X^T) X\beta = X \cancel{(X^T X)^T} X^T \beta = X\beta.$$

An important result for inference:

"The fitted values $\hat{y}$ & the residuals e are indep."

$$\text{i.e. } \hat{y} = X\hat{\beta} \perp e = y - \hat{y} = y - X\hat{\beta}$$

Aside: projection

Any projection matrix has only 0 or 1 eigenvals.

$$H = U \Lambda U^T$$

$$H = \underset{n \times n}{U} \underset{n \times n}{\Lambda} \underset{n \times n}{U}^T$$

$$H^2 = (H)(H) = (U \Lambda U^T)(U \Lambda U^T)$$

$$= U \Lambda \cancel{U^T U} \Lambda U^T = U \Lambda \Lambda U^T \\ = U \Lambda^2 U^T$$

eigenvals of H must only be those that satisfy $\lambda^2 = \lambda$

$$\Rightarrow \lambda = 0, \text{ or } 1$$

$$H = U \begin{bmatrix} \overset{p}{\overset{1}{1}} & & \\ & \ddots & \\ & & \overset{n-p}{1} \\ \hline & & & 0 & \dots & 0 \end{bmatrix} U^T$$

How many 1s?
 ~ ~ 0s?

$$\text{rank}(H) = \text{tr}(H) = \text{tr}(U \Lambda U^T) = \text{tr}(\Lambda U^T U) \\ \parallel \\ = \text{tr}(\Lambda) = p$$

$$\text{tr}(X(X^T X)^T X^T)$$

||

$$\text{tr}(X^T X)^T X^T X) \\ = \text{tr}(I_p) = p$$

$\Lambda \quad \Lambda$

$$H = U \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix} U^T$$

$$\text{Let } U = \begin{bmatrix} U_1 & U_2 \\ n \times p & n \times (n-p) \end{bmatrix}$$

$$H = \begin{bmatrix} U_1 & \underline{U_2} \end{bmatrix} \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} = U_1 U_1^T$$

Similarly,

$$I - H = I - U \Lambda U^T$$

$$= U U^T - U \Lambda U^T = U (I - \Lambda) U^T = U \begin{bmatrix} 0 & 0 \\ 0 & \underline{I_{n-p}} \end{bmatrix} U^T$$

$$= \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} p & 0 \\ 0 & I_{n-p} \end{bmatrix} \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix} = u_2 u_2^T //$$

$\therefore$

$$Y \sim N(X\beta, \sigma^2 I)$$

$$u^T Y \sim N(u^T X\beta, \sigma^2 I)$$

And $u^T = \begin{bmatrix} u_1^T \\ u_2^T \end{bmatrix}$ so,

$$u^T Y = \begin{bmatrix} \boxed{u_1^T Y} \\ \boxed{u_2^T Y} \end{bmatrix} \sim N \left(\begin{pmatrix} u_1^T X\beta \\ u_2^T X\beta \end{pmatrix}, \sigma^2 \begin{pmatrix} I_p & 0 \\ 0 & I_{n-p} \end{pmatrix} \right)$$

$\Rightarrow u_1^T Y \perp u_2^T Y$ b/c they're joint normal & their cov is 0.

$$u_1^T Y \perp u_2^T Y$$

$$u_1 u_1^T Y \perp u_2 u_2^T Y$$

$$HY \perp (I-H)Y$$

$$\hat{\gamma} \perp \underline{e}$$

This will allow us to do inference on β !

A rough look ahead to general F-tests... we will cover in a concise way later!

Why does this matter?

$$H_0: \beta = \beta^*$$

$$H_1: \beta \neq \beta^*$$

$$\hat{\beta} \sim N(\beta, \sigma^2(X^T X)^{-1})$$

$$\hat{\beta} - \beta \sim N(0, \sigma^2 \underline{(X^T X)^{-1}})$$

$$t := \frac{\bar{z}}{\sqrt{v/d}}$$

$$24 \sqrt{v}$$

$$\underline{(\sigma^2(X^T X)^{-1})}^{-1/2} (\hat{\beta} - \beta) \sim N(0, I)$$

||

$$\underline{(X^T X)^{1/2} (\hat{\beta} - \beta)}$$

σ

If I have independence of $\hat{\beta}$ & e ($\therefore \hat{\sigma}^2$)
then I can estimate σ by $\hat{\sigma}$ & the
resulting qty will follow a t_{n-p} dist.

Plug in $\hat{\sigma}$ for σ :

$$\frac{(X^T X)^{1/2} (\hat{\beta} - \beta)}{\hat{\sigma}}$$

...

?

$$\left\| \frac{(X^T X)^{1/2} (\hat{\beta} - \beta)}{\hat{\sigma}} \right\|^2 = \frac{\left[(X^T X)^{1/2} (\hat{\beta} - \beta) \right]^T \left[(X^T X)^{1/2} (\hat{\beta} - \beta) \right]}{\hat{\sigma}^2}$$

$$= \frac{(\hat{\beta} - \beta)^T (X^T X)^{1/2} (X^T X)^{1/2} (\hat{\beta} - \beta)}{\hat{\sigma}^2}$$

$$= \frac{(\hat{\beta} - \beta)^T (X^T X) (\hat{\beta} - \beta)}{\hat{\sigma}^2} \cdot \frac{1}{\sigma^2} \dots$$

eventually
this
idea will
lead to an
F-stat.
next wk!

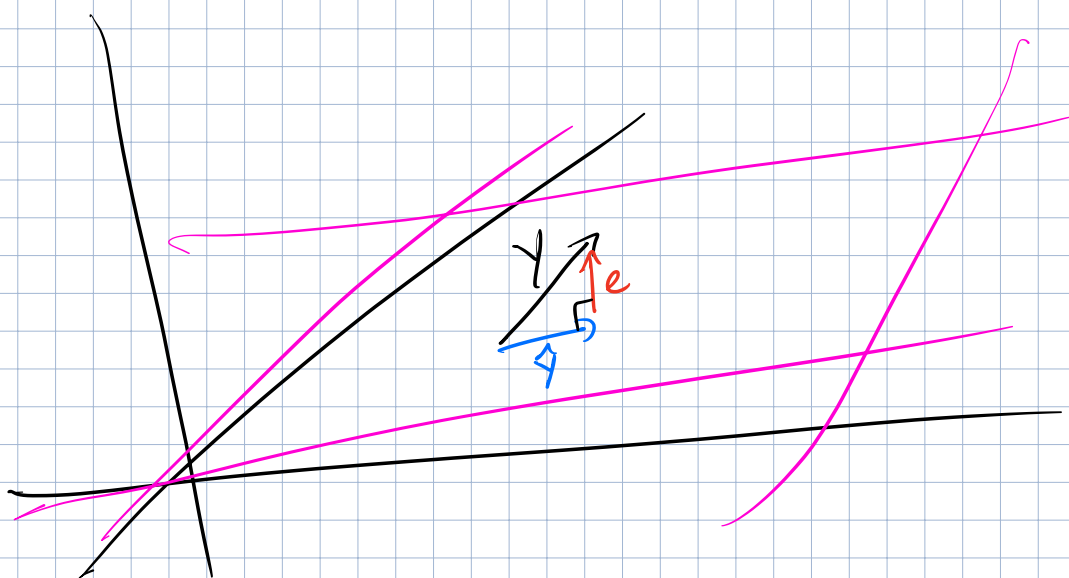
Note: $\left\| \frac{(X^T X)^{1/2} (\hat{\beta} - \beta)}{\sigma} \right\|^2 \sim \chi_p^2$

$\hat{\sigma}^2 = \frac{SSE}{\sigma^2(n-p)} \Rightarrow \frac{(n-p)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2$

$$\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$$

$$\frac{(X^T X)^{1/2} (\hat{\beta} - \beta)}{\sigma} \sim N(0, I_p)$$

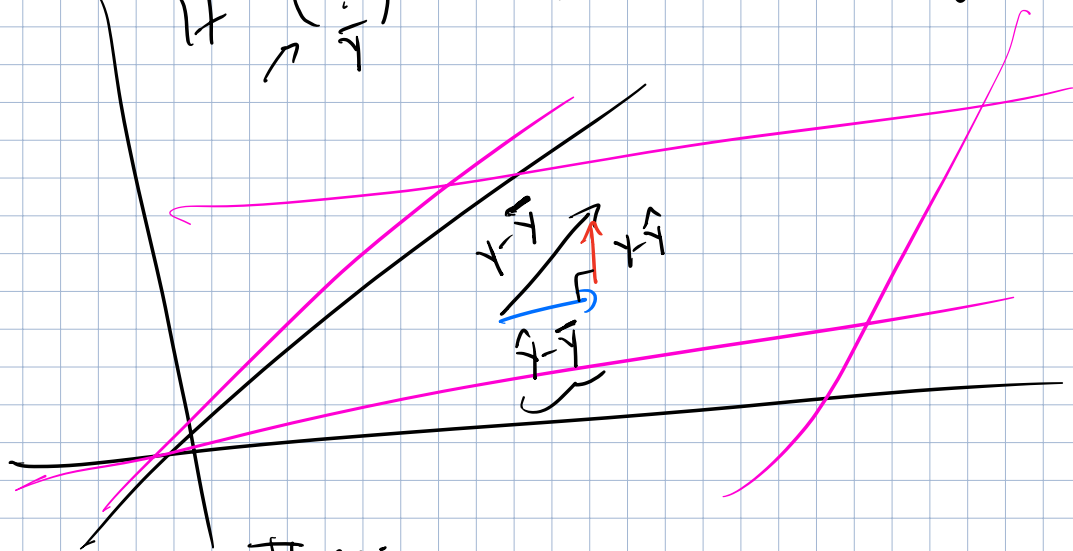
Geometry of Regression:



Prove that $\hat{y}^T \underline{e} = 0$

$$\begin{aligned}
 & (HY)^T (I-H)Y \\
 &= Y^T H^T (I-H)Y \\
 &= Y^T H (I-H)Y \\
 &= Y^T (H-H^2)Y \\
 &= Y^T (H-H)Y = Y^T 0Y = 0
 \end{aligned}$$

if $\begin{pmatrix} \bar{y} \\ \hat{y} \end{pmatrix} \in \text{Colspace}(X)$ (which is true if I have an intercept)



Pythagorean Thm:

$$\|y - \hat{y}\|^2 + \|\hat{y} - \bar{y}\|^2 = \|y - \bar{y}\|^2$$

$$SSE + SSR = SST$$